Integrated IoT-Based Sustainable Robotic System for Multifunctional Gardening Purposes

Aman Rao* Aman Gawar*, Digambar Kumar*, Himanshu Kumar*
*School of Computer Applications, Lovely Professional University,
Phagwara, Punjab, India

***Abstract*—This review article digs into the creation of a battery-powered robotic gardener that aims to deliver eco-friendly and automated plant care. One of the outstanding features of the work is the presence of numerous intelligent features that consist of automatic grass cutting, obstacle detection, soil moisture monitoring, and waste bin indication, all incorporated into a tiny Arduino-based robotic platform. The system utilizes ultrasonic sensors for obstacle detection to stop the cutting operation if a human or object is detected, thus ensuring safety during the operation. A soil moisture sensor constantly monitors the earth's humidity level and if it is found to be dry, watering follows. Some of the ways through which the system informs about the bin in which the cutting grass is collected and is full are LED indicators and a buzzer sound. The reviewed sources reveal the impressive development in the field of robotic and smart gardening systems that also entail solar-powered cutters and AI-based garden automation. The described robotic gardener has the device features such as the device's efficiency, safety, and sustainability for home and small-scale garden applications, and it is still powered by rechargeable batteries for uninterrupted performance.**

***Index Terms*—Robotic Gardener, Smart Irrigation, Automation, Arduino, Obstacle Detection, Soil Moisture Sensor, Renewable Energy**

I. Introduction

Gardening is essential for helping the environment by maintaining the balance of nature, giving people positive mental health, and making the environment more beautiful. However, with the rise of city life, manual gardening is becoming increasingly difficult due to the lack of time and the increase in maintenance costs. To solve these problems, automation and robotics have become popular and effective ways of keeping gardens with little human intervention. Robotic gardeners are created to carry out chores such as lawn mowing, watering, and plant monitoring without human intervention. They work effectively with the help of their sensors and controllers that guarantee accurate operations.

With the advent of Arduino-based robotic systems, smart gardening has become cheap and easy to reach for people living in their own houses. In old systems, lawn cutting and watering were done by hand and thus were time-consuming activities. Present-day robotic systems incorporate sensing technologies that allow them to monitor environmental parameters such as soil moisture, light intensity, and plant health in real-time.

The robotic gardener being introduced runs on batteries that can be recharged and is fitted with sensors that can identify soil moisture, grass height, and obstacles. It is also equipped with a bin alert system using LEDs and sound signals to notify the user when the collection bin is full. This review briefs the survey conducted on recent robotic gardening-solutions research and highlights current issues addressed by improvements made to sustain automation and ecology.

II. Literature Survey

Robotic gardening and automated lawn maintenance have been a focus of research for the last ten years. Scientists have energized different setups by solar energy, batteries and IoT to lower the hard work in labor and make the system more efficient. The following section is a review of five significant papers on lawn care and gardening robots, which discusses their methodologies, technologies, and scope of limitations.

J. Chincholkar et al. in 2016, invented a solar-powered lawn mower. The cutting of the grass done by the robot was up to 3 inches and the robot was also able to move on uneven ground and run under the light of the sun all by itself. The study of J. Pradeep et al. (2018) presented a solar grass cutter that was controlled by a mobile phone and used GPS for navigation and Bluetooth for remote control thus giving a good result in area of a small lawn. After that, T. Wong et al. (2020) came up with a solar grass cutter coupled with machine learning algorithms which helps the robot to automatically change its cutting height by the density of the grass. A. Verma (2022) introduced a battery-powered autonomous mower that used ultrasonic sensors for detection of obstacles and the control was in the hands of the Arduino board. In the end, M. Das (2023) talked about a smart robotic lawn mower which integrated soil moisture sensors along with battery power to make lawn care more eco-friendly. These scientists show how the concept of gardening robots has changed from solar-powered to AI-powered robots with sensors that are compact and battery-powered and suitable for small-scale applications.

III. Related Work

The development of smart and automated gardening systems has been a major focus of numerous research projects over the last ten years. Mechsy et al. created a mobile robot-based watering system that irrigation automatically irrigates a lawn using a soil moisture sensor, the point being water efficiency and reduction of human labor. Similarly, Pullannagari et al. introduced an IoT-driven solar-powered robotic mower and

TABLE I
COMPREHENSIVE LITERATURE REVIEW ON ROBOTIC GARDENING AND SMART LAWN MAINTENANCE SYSTEMS

S. No.	Author(s) & Year	Focus Area / System Description	Technology / Algorithm Used	Key Outcomes / Limitations
1	Mechsy et al. (2017) [1]	Mobile robot for automated lawn watering	Microcontroller-based control, sensors	Efficient watering mechanism but lacks obstacle avoidance.
2	Pullannagari et al. (2025) [2]	IoT-driven solar robotic mower and sprinkler	IoT, Solar Power, Embedded Systems	Combines mowing and watering; needs better terrain adaptability.
3	Arora et al. (2019) [3]	Autonomous lawn mower with water sprinkler	DC Motors, Solar Power, Microcontroller	Provides dual functionality; lacks AI-based adaptability.
4	Gravalos et al. (2019) [4]	Robotic irrigation for urban gardens	Sensors, Actuators, IoT Framework	Efficient irrigation; limited to urban environment scale.
5	Verne (2020) [5]	Human-robot adaptation for lawn mowers	HRI Study, User Interaction Models	Addresses human adaptation; not focused on automation.
6	Ramya et al. (2021) [6]	Solar grass cutter with spraying system	Arduino, Solar Power, Relay Module	Reduces manual effort but limited in real-time sensing.
7	Rajmani et al. (2019) [7]	Hybrid operating grass cutter	Battery + Solar, Mechanical Design	Sustainable dual power; lacks automation.
8	Ismail et al. (2019) [8]	Smart solar grass cutter	IoT, Solar Panel, Arduino	Cost-effective and efficient; no obstacle detection.
9	Dilip et al. (2017) [9]	Automatic solar grass cutter design	DC Motors, Solar Power, Sensors	Automates cutting; no moisture or bin-level sensing.
10	Ramisetti et al. (2024) [10]	Arduino-based grass cutting robot	Ultrasonic Sensor, Arduino, DC Motor	Real-time obstacle avoidance; lacks irrigation integration.
11	Man et al. (2022) [11]	Dual-mode grass cutter robot	Dual motor drive, Battery, Sensor-based control	Manual and autonomous modes; moderate accuracy.
12	Aditya Rajmani et al. (2019) [**?**]	Hybrid operating grass cutter	Battery + Solar Combination	Good energy efficiency; lacks IoT connectivity.
13	Neha et al. (2018) [13]	IoT-based automated grass cutter	IoT, Wi-Fi Module, Arduino	Enables remote monitoring; limited power optimization.
14	Prashanth et al. (2024) [14]	Automated solar grass cutter	Solar Power, Arduino, Sensor Control	Simple and efficient; not adaptable to uneven terrain.
15	Asha et al. (2017) [15]	Smart grass cutter system	Microcontroller, DC Motor, Sensor	Compact and cost-effective; lacks multi-sensor fusion.

sprinkler system, thus combining energy efficiency and environmental sustainability with garden automation.

Arora et al. built an autonomous lawn mower with a water sprinkler that simultaneously handled grass cutting and irrigation. Gravalos et al. took this idea further by creating a robotic irrigation system for urban gardening that uses sensors to achieve water efficiency. Ramya et al. came up with a solar-powered grass cutter that could spray water, combining two eco-friendly concepts - energy sustainability and lawn coverage.

Besides, Verne and Ismail et al. provided insights into the user-friendliness of robotic mowers and their adaptability in real-life scenarios, while also addressing ergonomic and design challenges. The works of Ramisetti et al. and Man et al. that followed were deeply involved with Arduino and ultrasonic sensors-based system for obstacle detection and dual-mode control, thus elevating automation safety and precision. In sum, these researches reveal the transition to intelligent, IoT-enabled, and energy-efficient robotic gardening systems that pledge sustainability, accuracy, and the least possible human intervention.

IV. SUGGESTIVE APPROACHES

The proposed battery-powered robotic gardener demonstrates an effective solution for sustainable and autonomous garden maintenance. However, further advancements can enhance its efficiency, adaptability, and intelligence. The following suggestive approaches aim to improve system performance, control precision, and automation intelligence for future implementations.

- **AI-Based Vision System:** Integrate a camera module with image processing and deep learning models (e.g., CNN) to identify plant species, detect weed growth, and assess plant health for selective care.
- **Wireless Communication:** Incorporate Wi-Fi or IoT-based modules (e.g., ESP8266 or NodeMCU) for remote monitoring, data logging, and real-time control through a mobile or web application.
- **Path Planning and Localization:** Implement GPS or SLAM (Simultaneous Localization and Mapping) techniques to enable the robot to navigate efficiently across larger gardens with predefined paths.
- **Energy Optimization:** Replace standard batteries with solar-assisted hybrid power units to extend operational time and reduce dependency on charging.

- **Smart Irrigation Scheduling:** Combine soil moisture readings with weather prediction data using an ML model to make intelligent watering decisions, optimizing water usage.
- **Enhanced Safety Mechanisms:** Introduce machine learning-based obstacle prediction and emergency shutdown systems for safer operation around humans and pets.
- **Modular Hardware Design:** Develop a modular frame allowing easy upgrades and maintenance, such as detachable mower units or irrigation nozzles.

TABLE II
SUGGESTED ENHANCEMENTS FOR ROBOTIC GARDENER SYSTEM

Improvement Area	Technology / Method	Expected Benefit
Plant Health Monitoring	AI-based Image Recognition	Early disease detection and targeted maintenance
Remote Control	IoT (Wi-Fi / Cloud)	Real-time monitoring and command execution
Navigation	GPS / SLAM	Efficient movement and path accuracy
Power Management	Solar-Assisted Battery System	Increased runtime and energy sustainability
Irrigation Control	ML-based Weather Prediction	Smart watering and reduced wastage
Safety	AI-driven Obstacle Detection	Enhanced operational safety
Hardware Modularity	Modular Frame Design	Ease of upgrades and maintenance

V. COMPARISON AND CHALLENGES

The new battery-powered robotic gardener system can be a major winning point compared to traditional and solar-powered robotic systems. In contrast to an earlier one which is entirely solar-powered or manually controlled, this system integrates a battery-based mobility, smart irrigation, and multi-sensor automation to realize performance, flexibility, and sustainability at a higher level. Also, its modular design and inexpensive components make it suitable for small-scale home and agricultural applications.

Nevertheless, the issue of adaptability to the environment, limited processing power, and lack of real-time intelligence are still some of the obstacles that the team has to overcome with this product. Eventually, the use of advanced AI techniques, wireless communication, and energy-efficient computing architectures can make the robotic gardener more reliable and efficient.

Key Challenges

- **Limited Autonomy:** The system lacks adaptive learning to adjust cutting or watering strategies dynamically.
- **Battery Efficiency:** Prolonged operation reduces power efficiency; hybrid solar-battery integration is needed.
- **Terrain Navigation:** Uneven surfaces or tall grass may hinder wheel traction and movement.

TABLE III
COMPARISON OF PROPOSED SYSTEM WITH EXISTING ROBOTIC GARDENING SOLUTIONS

Feature	Existing Systems (Solar-Based)	IoT-Enabled Robots	Proposed Battery-Powered System
Power Source	Solar energy only	Solar + Wired	Rechargeable Battery (Portable)
Control Type	Manual / Semi-auto	IoT-based remote control	Fully automated with Arduino
Sensor Integration	Limited (obstacle only)	Multi-sensor setup (temp, humidity)	Multi-sensor + moisture, obstacle, bin detection
Mobility	Low torque motors	Medium speed	High torque geared motors for smooth movement
Irrigation Capability	Absent or manual	Remote-based activation	Automated water pump via soil moisture sensor
Intelligence Level	Rule-based	Cloud-monitored	AI-Ready (Upgradeable)
Maintenance	Moderate	High (complex IoT setup)	Low (simple modular hardware)

- **Real-Time Monitoring:** Absence of IoT/cloud integration limits remote supervision and performance analysis.
- **Environmental Durability:** Exposure to dust, water, and temperature variations can affect sensor performance.

TABLE IV
SUMMARY OF REVIEWED SYSTEMS

Author & Year	System Description	Technology Used	Key Outcomes
J. Chincholkar et al. (2016)	Solar-powered grass cutter with rechargeable battery.	Solar Panel, DC Motor, Blades, Rechargeable Battery	Effective grass cutting up to 3 inches height.
J. Pradeep et al. (2018)	Mobile-controlled solar grass cutter with GPS.	GPS, Bluetooth, DC Motors, Solar Power	Remote navigation and improved precision cutting.
T. Wong et al. (2020)	ML-based solar cutter with height adjustment.	Sensors, ML Algorithm, Solar Energy	Adaptive cutting based on grass density.
A. Verma (2022)	Autonomous mower using ultrasonic sensors.	Arduino, Ultrasonic Sensor, Battery, DC Motor	Enhanced safety through obstacle detection.
M. Das (2023)	AI-assisted mower with soil sensors.	Microcontroller, Soil Moisture Sensor, DC Motor, Battery	Sustainable and efficient garden automation.

VI. CONCLUSION

Creating a robotic gardener that operates on battery power is a major step forward for environmentally friendly and

automated garden care Chincholkar et al., 2016; Verma, 2022; Das, 2023. This article traces the changes in garden robots, which are now AI supported and sensor equipped, from merely solar powered lawn mowers. The suggested scenario with soil moisture monitoring, automatic cutting, obstacle detection, and bin level alerts significantly decreases manual intervention while keeping the garden in a fine state of health Wong, 2020; Pradeep, 2018.

The project will also be safe since the ultrasonic sensors will be used to ensure that the robot does not come into contact with people or objects. Moreover, figures measured by the soil moisture sensors will be used only for watering, thus saving resources. Portable rechargeable batteries are also environmentally friendly, and modular design makes it easier to maintain and upgrade in the future Ramisetti, 2024; Ramya, 2021.

In future, the system may take advantages of AI powered computer vision to keep track of plant health, recognize weeds, and measure the rate of growth besides that, it may use wireless technology to be controlled and monitored from a distance Pullannagari, 2025; Arora, 2019. These changes could allow the device to be used in bigger gardens or as part of a smart home system. To sum it up, the robotic gardener presented here is an illustration of how IoT, automation, and green technology may be combined to provide a modern, effective, safe, and environmentally friendly garden management solution.

REFERENCES

[1] L. S. R. Mechsy, M. U. B. Dias, W. Pragithmukar, and A. L. Kulasekera, ”A mobile robot based watering system for smart lawn maintenance,” in *2017 17th International Conference on Control, Automation and Systems (ICCAS)*, Oct. 2017, pp. 1537-1542, IEEE.

[2] P. Pullannagari, B. Penmetsa, N. Sappa, H. Myle, and A. Naureen, ”IoT-driven solar-powered robotic lawn mower and water sprinkler,” in *Recent Trends in VLSI and Semiconductor Packaging*, 2025, pp. 52-59, CRC Press.

[3] H. Arora, J. A. Sagor, V. Panwar, P. Sharma, S. K. Mishra, P. G. Arora, and P. K. Singh, ”Design and fabrication of autonomous lawn mower with water sprinkler,” *Think India Journal*, vol. 22, no. 17, pp. 2101, 2019.

[4] I. Gravalos, A. Avgousti, T. Gialamas, N. Alfieris, and G. Paschalidis, ”A robotic irrigation system for urban gardening and agriculture,” *Journal of Agricultural Engineering*, vol. 50, no. 4, pp. 198-207, 2019.

[5] G. B. Verne, ”Adapting to a robot: Adapting gardening and the garden to fit a robot lawn mower,” in *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, Mar. 2020, pp. 34-42.

[6] E. Ramya, J. Anand, R. R. Devi, N. A. Issac, and K. R. Prasenth, ”Solar grass cutter with water spraying vehicle,” in *2021 International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)*, Oct. 2021, pp. 1-6, IEEE.

[7] A. S. Rajmani, A. N. Gaonkar, A. Darak, A. Joshi, and V. M. Murgod, ”Design and Fabrication of Hybrid Operating Grass Cutter,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 8, no. 5, 2019.

[8] F. B. Ismail, N. F. Al-Muhsen, A. F. Fazreen, and A. Zukipli, ”Design and development of smart solar grass cutter,” *International Journal of Engineering and Advanced Technology*, vol. 9, no. 2, pp. 4137-4141, 2019.

[9] B. P. Dilip, N. B. P., V. S. U., S. W., and P. S. M., ”Design and Implementation of Automatic Solar Grass Cutter,” *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, vol. 6, no. 4, 2017. [Online]. Available: http://www.ijareeie.com/volume6-issue-4

[10] V. Ramisetti, P. S. Chowdary, C. Bankuru, S. Peda, and M. Pasala, ”Automatic grass cutting robot using Arduino and ultrasonic sensor,” in *2024 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, Mar. 2024, pp. 1-4, IEEE.

[11] A. E. C. Man, S. R. Aw, N. Abd Rahman, R. S. N. A. R. Aris, L. S. Syafie, M. H. Othman, et al., ”Development of Dual-Mode Grass Cutter Robot,” *Journal of Physics: Conference Series*, vol. 2319, no. 1, p. 012012, Aug. 2022.

[12] A. S. Rajmani, A. N. Gaonkar, A. Darak, A. Joshi, and V. M. Murgod, ”Design and Fabrication of Hybrid Operating Grass Cutter,” *IJERT*, vol. 8, no. 5, pp. 795-799, May 2019.

[13] S. A. Neha, ”Automated Grass Cutter Robot Based on IoT,” *International Journal of Trend in Scientific Research and Development*, vol. 2, no. 5, pp. 334-337, Aug. 2018.

[14] K. Prashanth, D. K. Yadav, M. D. Safi, and T. Praveen, ”Automated Solar Grass Cutter,” 2024.

[15] N. Asha, J. S. Monica, R. Saraswathi, and R. Rahul, ”Smart grass cutter,” *Perspectives in Communication, Embedded-systems and Signal-processing-PiCES*, vol. 1, no. 6, pp. 97-99, 2017.